

ON AN APPROACH TO OPTIMIZING SOLUTIONS TO INFORMATION SEARCH PROBLEMS IN CYBER PHYSICAL SYSTEMS. PART 1. REDUCING THE OPERATION TO I-EQUIVALENT

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1 Abstract

The article presents a general approach to optimizing solutions to information retrieval problems in cyber physical systems for various purposes in the context of Big Data. The relevance lies in the low efficiency of traditional methods used today by companies focused on selling goods when making informed decisions. It is proposed to consider them from the perspective of the requirements for cyber-physical systems with the reduction of information retrieval operations to the I-equivalent. A formalized formulation of the problem and modeling of the information search process of a cyber-physical system was carried out in compliance with the “feedback” principle. The reduction of the information search process to the I-equivalent is shown. An option for implementing the proposed approach (technology) for a separate company has been developed.

2 Keywords

Cyber physical system, Big Data, decision maker, metric connectivity, triplet, information equivalent (I-equivalent), descriptor, denotation, marker, encapsulation, computer network.

3 Introduction

Scientific and technological progress at the present historical stage has acquired the character of a factor of global importance. In our time, it largely determines the shape of the world economy, world trade, and relations between states and regions. The most significant contribution of scientific and technological

progress today is observed in computer technology (Figure 1). Thus, over the past 50 years, there has been a rapid development of computer technology from the typewriter to the personal computer. The amount of data stored online is increasing by approximately 40% annually.

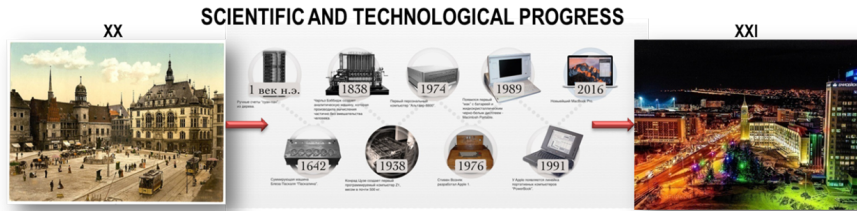


Figure 1: Scientific and technological progress

Global informatization of all spheres of society has led to the unification of heterogeneous information into large data sets - BIG DATA. Today, a problematic situation has arisen when the volume of circulating information exceeds the ability of modern companies to evaluate it (Figure 2). The ratio of the analyzed options to the total number does not allow making an informed decision. A number of factors influencing events are beyond the limits of the calculated sample. Which leads to an unpredictable development of the situation (Figure 3) [3]. The solution to this problem lies in the plane of the concept of Big



Figure 2: Informatization of society and Big Data

Data, a description of the operation of the system generating Big Data and the definition of the principles for working with it by cyber physical systems[1].

Big Data includes structured and unstructured data that is efficiently processed by horizontally scalable software tools and, as a result, makes rational decisions [2].

Based on this, the problem can be decomposed into 2 parts (Figure 4):

Quality. Consists in heterogeneous data syntax. To identify semantics, their normalization is required.

Quantity. Consisting of a large amount of data. To process them, it is necessary to ensure performance corresponding to the time of their relevance. Global information network generating Big Data is a self-organizing system (or



Figure 3: Risks for modern company

dynamic) adaptive system in which memorizing information (accumulation of experience) is expressed in changing the structure of the system.

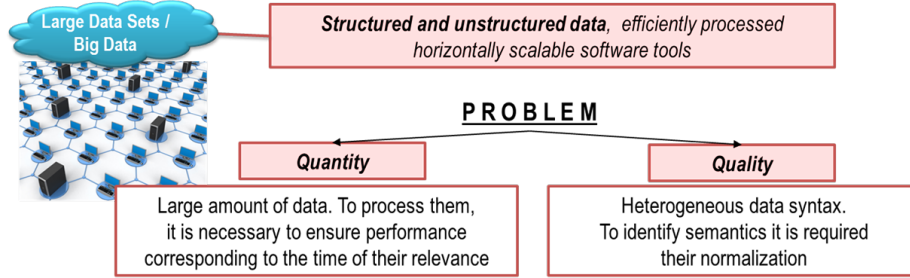


Figure 4: Big Data problem decomposition

Cyber Physical system - information technology concept. Computational components are distributed throughout the physical system, which is its carrier, and synergistically linked with its constituent elements. Based on this, modern cyber physical systems can include organizations from commercial firms to industrial enterprises that integrate physical components (sensors, devices) with computing processes to interact with their environment and perform on-line tasks. Cyber physical systems are characterized by the laws and regularities of cybernetics. The qualitative part of the problem is solved in cybernetics by observing the feedback principle [4]. The essence of the feedback principle is that without feedback between elements, parts or systems, it is impossible to organize their effective management. Feedback signals the achieved result. Based on this information, the control action is adjusted. Implementation of this principle in the operation of a cyber physical system with Big Data.

4 Mathematical formulation of the problem

Metric transformation of the function describing the variable $f(\cdot)$ **on the forward loop**, removing uncertainty by “imposing” the function of describing the variable $f(\cdot)$ on the field of options $\tilde{?}$, metric transformation of the function

describing a variable with its value $f(Y)$ **on the return loop**. Comparison of the variable description function with an attached reverse loop value and the forward loop variable description function in the decision maker (DM) metric. **The goal** is to achieve equality ($f_m(\tilde{m}) = f^*m(Y_m)$).

On the forward loop (Figure 5)

1. The decision maker formulates *the initial request* $\mathbf{f}$ (problem statement) according to the description rule $\mathbf{m}$ specified by him.
2. The connection rule between description $\mathbf{m}$ and node 1 ($\mathbf{L}$) is specified.
3. The request $\mathbf{f}$ generated according to rule $\mathbf{m}$ is transmitted to node 1 with its transformation according to the rule for describing node 1 ($\mathbf{L}$).
4. The connection rule for the description $\mathbf{L}$ of node 1 and the rules for describing nodes 2 ($\mathbf{s}$) and 3 ($\mathbf{q}$) are specified, respectively.
5. A request $\mathbf{f}$ generated according to rule $\mathbf{L}$ is transmitted to nodes 2 and 3 with a complex transformation of it according to the rules for describing nodes $\mathbf{s}$ and $\mathbf{q}$.
6. Next, similar transformations are performed with the transmission of the request according to the rules $\mathbf{g}$, $\mathbf{h}$, $\mathbf{r}$, $\mathbf{t}$ (sensors).
7. To the node for solving the problem (removing uncertainty).

Normalization of data syntax is carried out by bringing it to a unified format for representing a semantic unit using mathematical operators: (differentiation, integration, convolution with the kernel, Fourier transform, Lie bracket).

ON THE FORWARD LOOP

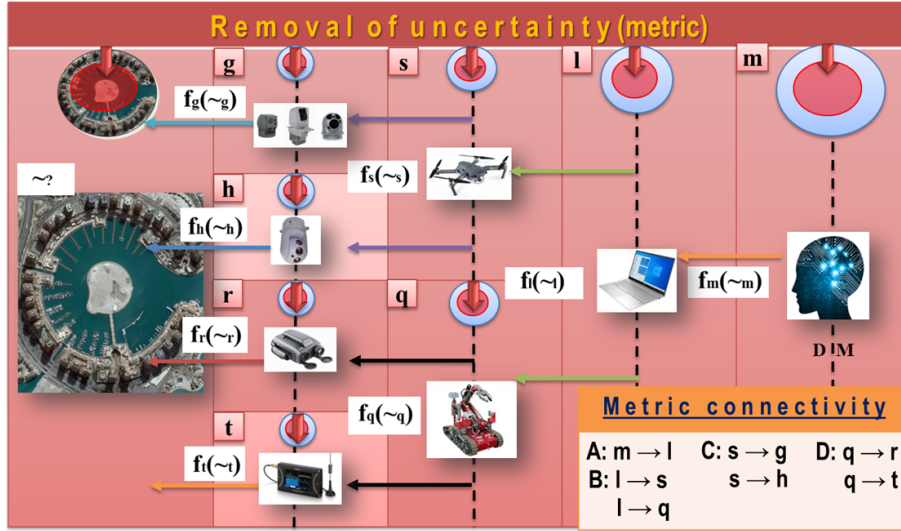


Figure 5: Forward loop feedback principle

On the return loop (Figure 6)

1. At the problem solution node, the uncertainty $f(\tilde{y})/\tilde{y} = y$.

2. The solution is formulated in the form of a complex answer f^* based on the metrics of sensors g, h, r, t .
3. According to a given connection rule, the response f^* generated according to the rules g, h, r, t is transmitted to nodes 2 and 3 with its transformation according to the rules for describing nodes s and q .
4. Next, a similar transmission with the transformation of the answer f^* according to the rule l .
5. The solution to the problem according to the description rule m specified by him is communicated to the decision maker in the form of a response to the initial request f^* . The condition for compliance with the principle is

ON THE RETURN LOOP

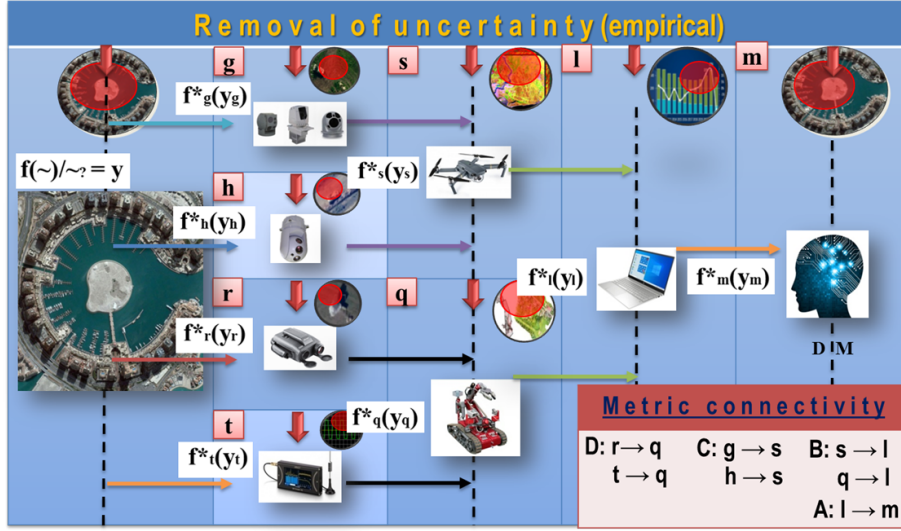


Figure 6: Return loop feedback principle

to ensure metric coherence between the description rules when transforming a query [6]. The work of mathematical transformations is based on **3 logical operators (triplet)** (disjunction, conjunction and inversion) [13], the practical implementation of which is carried out by specifying an encoding algorithm and an encapsulation model when transforming and transmitting *a request* between nodes [7]. This process can be implemented in information and telecommunication networks, ranging from peer-to-peer networks to more complex ones (with a dedicated server, camping networks, etc.), built on the principles of repeated data processing *with a comparison of the values of the input request (on a direct loop) and the output response (on the return loop) of the control cycle (DDM - data driven method)* [8]. Based on this, the qualitative side of the Big Data problem can be resolved by implementing *an approach that allows processing unstructured data* in information and telecommunication networks

(information (computer) networks) *with a comparison of the values* of the input request (on the forward loop) and the output response (on the reverse loop) of the cycle management, i.e. bringing the process to an **information equivalent (I-equivalent)**.

5 Solution option developed for the marketing department

This approach is implemented as a solution option developed for the marketing department [15]. *The essence of the developed solution* is to build a process for collecting data about a product on the market using specified analysis markers, with the conversion of the request by software applications and providing service connectivity between them when transferred from node to node of the information retrieval system of a marketing company [14].

The transfer is carried out according to the principle without loss of markers along the entire data track to the product, with verification of the identity of the collected data markers to the request markers at each reverse node [11].

A typical marketing department staff may include the following specialists:

- **head of the marketing department (marketing director, creative director)**. Coordinates the work of all employees, participates, if possible, in the development and launch of projects;
 - **marketing manager**. Responsible for specific marketing campaigns and projects, including the development of promotional materials, promotions and events;
 - **analyst**. Collects and analyzes data about the market, competitors and consumers, compiles reports and forecasts;
 - **product marketer**. Responsible for the promotion of specific products and/or product lines, develops marketing strategies;
 - **PR specialist**. Engages in public relations, interacts with the media and builds a positive image of the company;
 - **content marketer**. Selects communication channels with the audience, develops content strategies, participates in the development of content materials;
- Other work of the marketing department is outsourced.

The process includes 10 main stages:

Stage 1 setting a task by the head of the company (marketing department, marketer) to collect data about the product on the market. The task can be posed verbally (verbally) or non-verbally (graphically).

Using a software application, a voice (image) is transcribed and converted into a text document (managers workplace, example - the **Transcribe** utility).

Stage 2 identification of analysis markers and their decomposition into private markers that form sufficient semantics of the task, based on the rule for formulating the initial request by the manager.

Using the context analyzer, a component analysis of the text is carried out, all sememes and semes of marker words up to the fundamental one (**descriptors**)

are identified from the semantic dictionary; if inaccuracies in denotation (interpretation) are identified, the analyzer fixes the layout according to the last correct result. For each descriptor, a semantic graph is formed (workplace of a marketing manager).

Stage 3 parallelization of private queries for identified descriptors due to the distributed architecture of the information retrieval system (Figure 7).

Using a spreadsheet, descriptors are converted into a relational form (for example, using **PyQt**), where each of them is assigned the identifier of the corresponding node for further processing of the request (workplace of the marketing manager).

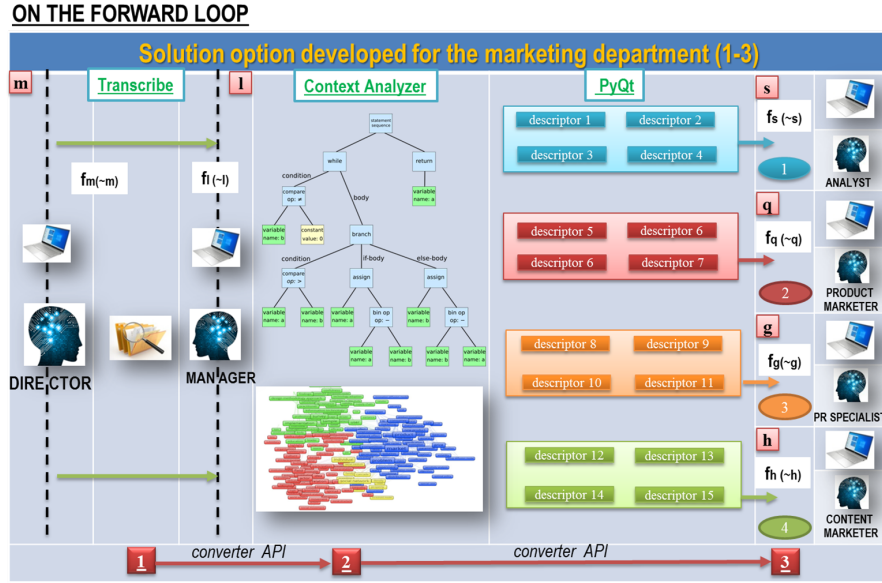


Figure 7: Solution option developed for the marketing department (1-3)

Stage 4 constructing a relevant request for each **descriptor** on each distributed node for transmission to an external browser.

Using SEO optimization applications [12], meta tags are created based on descriptors, semantic URL queries are generated based on them, and queries are checked for relevance indexes to descriptors (workplaces for analyst, product marketer, PR specialist, content marketer, example - using **Almaz BI** and **PyQt**).

Stage 5 creation and optimization of target URL requests by types of Big Data sources.

Using a mining tool, based on the semantic graph of the descriptor underlying each URL request, a tree of sources is formed according to the criterion **context-descriptor graph expansion**. Sources (social networks, email, instant messengers, hosting, blogs, forums, websites, Internet of things, etc.) are

ranked according to the depth of the graph disclosure. A table of priorities of target URL requests by source type is compiled. This is done by adding appropriate extensions to target URL requests (marketing department server, example - using **Hadoop, Pandas, Spark**) [5].

Requests in sequence from the highest rank (priority table) are transmitted to the generated request addresses (for example, requests are entered directly into **Safari**) (Figure 8).

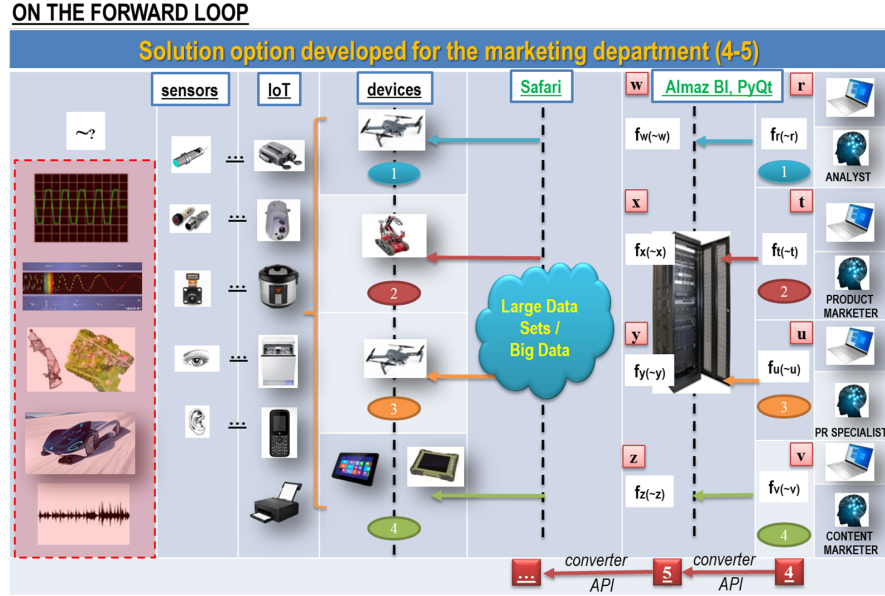


Figure 8: Solution option developed for the marketing department (4-5)

Stage 6 registration of search heuristics based on the minimax principle. The data that most deeply reveals the descriptor graph is recorded. Duplicate data, as well as data with lower ranks, are discarded. The data is matched to relevant Stage 4 queries and sent to each source node (marketing server, example using **DataLake**).

Stage 7 reverse engineering of data received by nodes based on relevant requests. Relevant query data is matched with meta tags created using SEO optimization applications [9]. Meta tags are converted into source **descriptors** to which the received data is attached (workplaces of an analyst, product marketer, PR specialist, content marketer, for example using **DataWarehouses, Statistica, Pandas**) (Figure 9).

Stage 8 comparison of data descriptors with the semantic graph of stage 2. Using the context analyzer, the structure of the semantic graph of the text is compared (according to the semantic dictionary, the sememe and seme of the marker words to the fundamental one (descriptors), if inaccuracies in the

ON THE RETURN LOOP

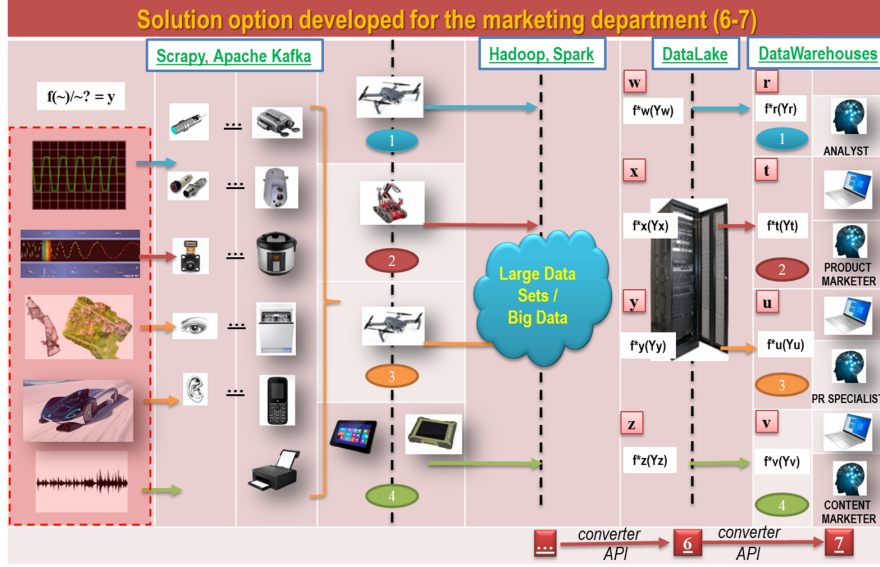


Figure 9: Solution option developed for the marketing department (6-7)

denotation (interpretation) are identified, the analyzer compares data for the last correct result. The comparison results are entered into special case lists. The case lists reflect the semantic graph of the original request and the received data for each position of the graph (workplace of a marketing manager, example using **PyQt**).

Stage 9 combining case lists with data into an initial array that forms the semantics of the problem. Converting an array of task semantics into a text response, taking into account the metric markers of the original request (structure, presentation style). Preparation of a visualization response according to the requirements specified in the task of the head of the company (marketing department, marketer) (use of graphs, diagrams, illustrations, photographs, animations) (workplace of the marketing manager, example - using **Matplotlib, Dash**).

Stage 10 bringing to the head of the company (marketing department, marketer) the answer to the initial request (task) (Figure 10).

Using a software application, speech is synthesized from a text response and a voice response is broadcast. A visualization response and a text response are demonstrated, taking into account the metric markers of the original request (managers workplace, example - using **Transcribe**).

When the manager generates the next target request, the process is repeated [10].

ON THE RETURN LOOP

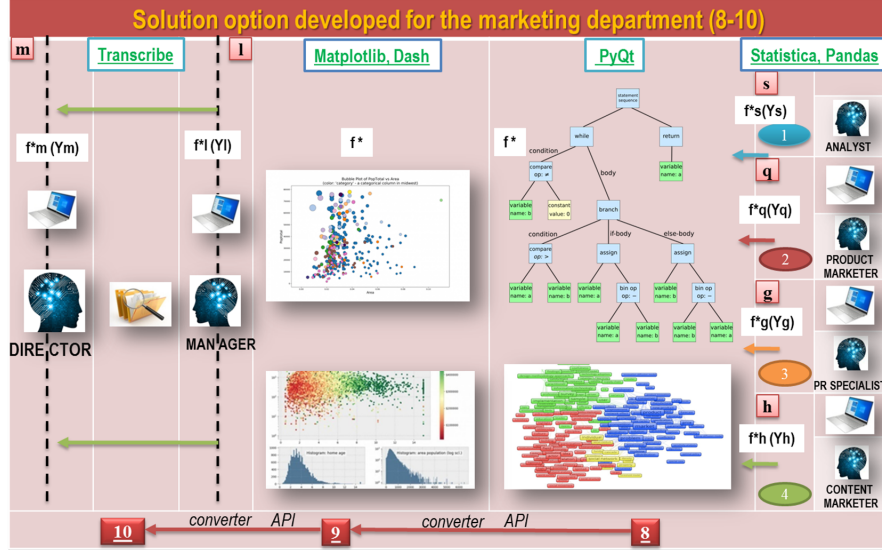


Figure 10: Solution option developed for the marketing department (8-10)

6 Conclusions

The developed general approach to optimizing solutions to data management problems in cyber-physical systems for various purposes in the context of big data allows us to consider the process of information retrieval with the reduction of operations to the I-equivalent. The formalized formulation of the problem and the modeling of this process should be carried out in accordance with the principle of “feedback”, and reduction to the I-equivalent through the implementation of the triplet coding model. Based on this approach, a business company’s technology can be designed (marketing department is an option) with specific software and hardware solutions that optimize the process of data collection, processing and analysis.

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